

Intro to Qt Quick

Overview

QML is a high level, scripted language. Its commands (known as elements) leverage the power and efficiency of the Qt libraries to make easy to use commands that perform intuitive functions. Draw a rectangle; display an image at a position and so on. Behind these elements are complex C++ libraries that efficiently perform the action.

The language also allows more flexibility of these commands by using Javascript rather than C++ to add new layers of logic to your application. Javascript is easier to learn than C++ and can be embedded into the QML files or imported from a separate file.

In QML the types of various 'objects' are referred to as elements.

An element usually has various properties that help define the element. For example, if we created an element called Circle then the radius of the circle would be a property.

A First Look

The basic syntax of an element is

- `SomeElement {`
- `id: myObject`
- `... some other things here ...`
- `}`

Here we are defining a new object. We specify its 'type' first as `SomeElement`. Then within matching braces `{ ... }` we specify the various parts of our element.

The id is a unique identifier for the element, it must start with a lower case letter and only contain letters, numbers and underscores. It is this particular object's name. If this `SomeElement` element was a `Rectangle` instead and it was one of many then the optional unique id would allow us to manipulate each element individually. Each visual element is ultimately based on, or inherits from, an element called `Item`. `Item` has certain properties and actions that may be useful.

Take a simple element such as a `Rectangle`. It has an id, we will call it `myRectangle`, it has a width and a height. Imagine that we want a rectangle that is 500 pixels by 400 pixels in the x and y directions (horizontal by vertical).

We can implement this `Rectangle` with these properties this way